

MEASURING WHEEL SYSTEMS

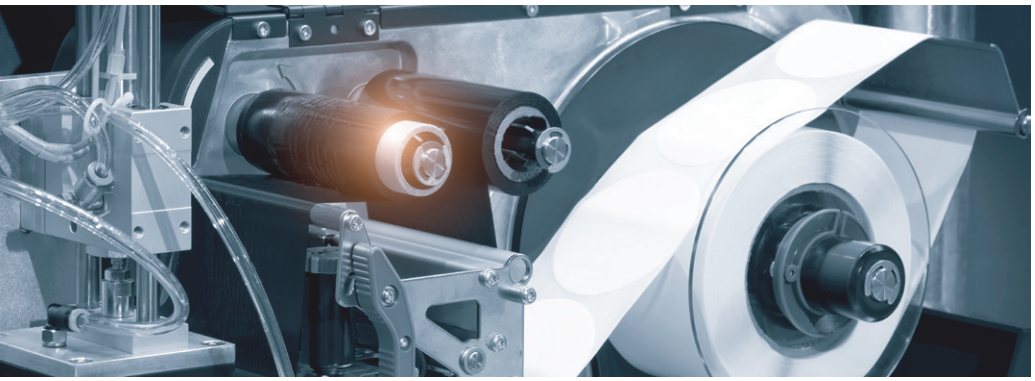
Speed measurement
Position detection
Length measurement

pulses for automation

Measuring wheel systems

Systems for speed measurement, position detection and length measurement

Measuring wheel systems from Kübler are the ideal solution for reliable speed measurement, position detection and length measurement in applications with linear movements. These are recorded rotationally via the measuring wheel with attached encoder directly on the surface of the material to be measured and converted into linear data. Integrated springs ensure the necessary contact force of the measuring wheel on the measuring surface for reliable measured value acquisition.





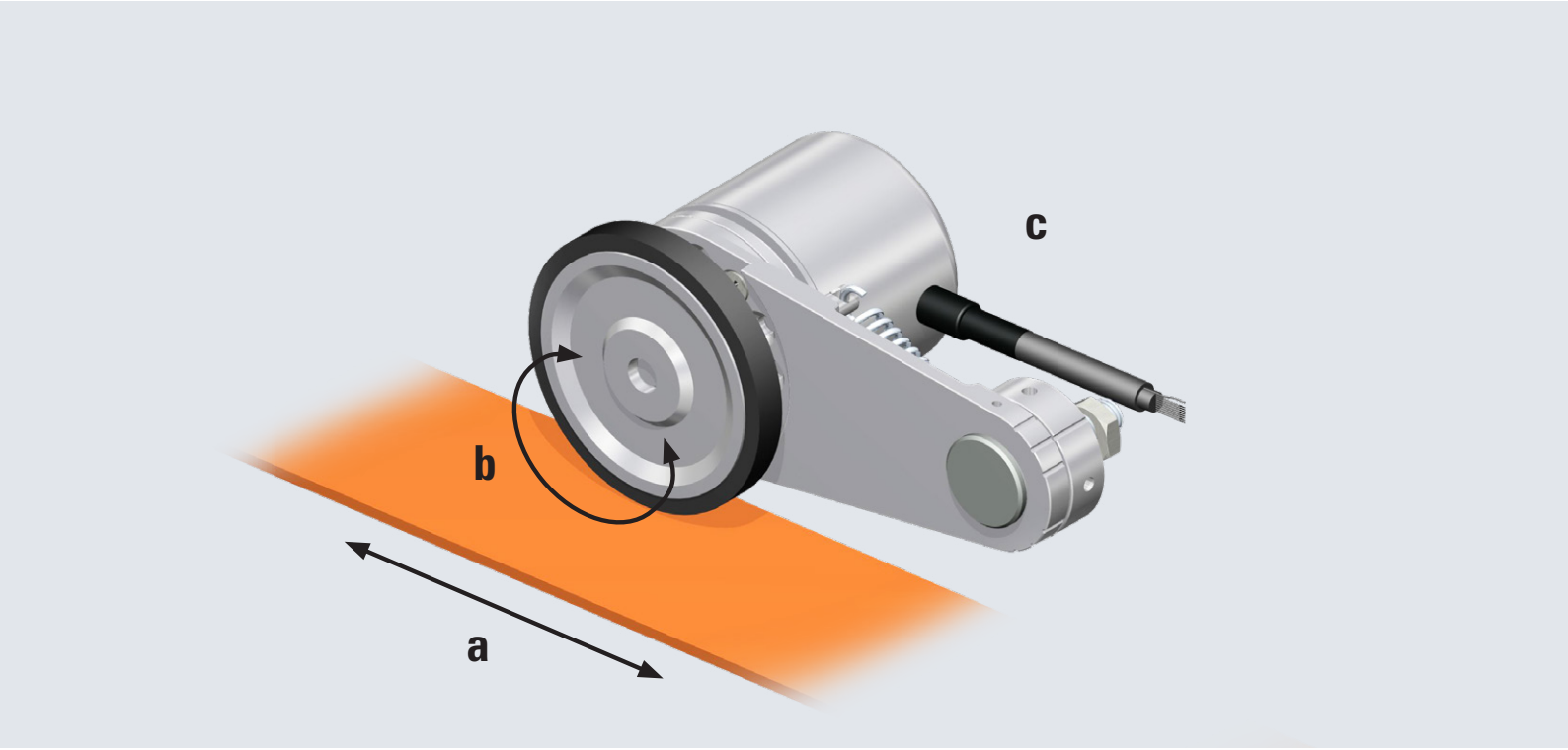


Further information on measuring wheel systems

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Measuring wheel systems - operating principle



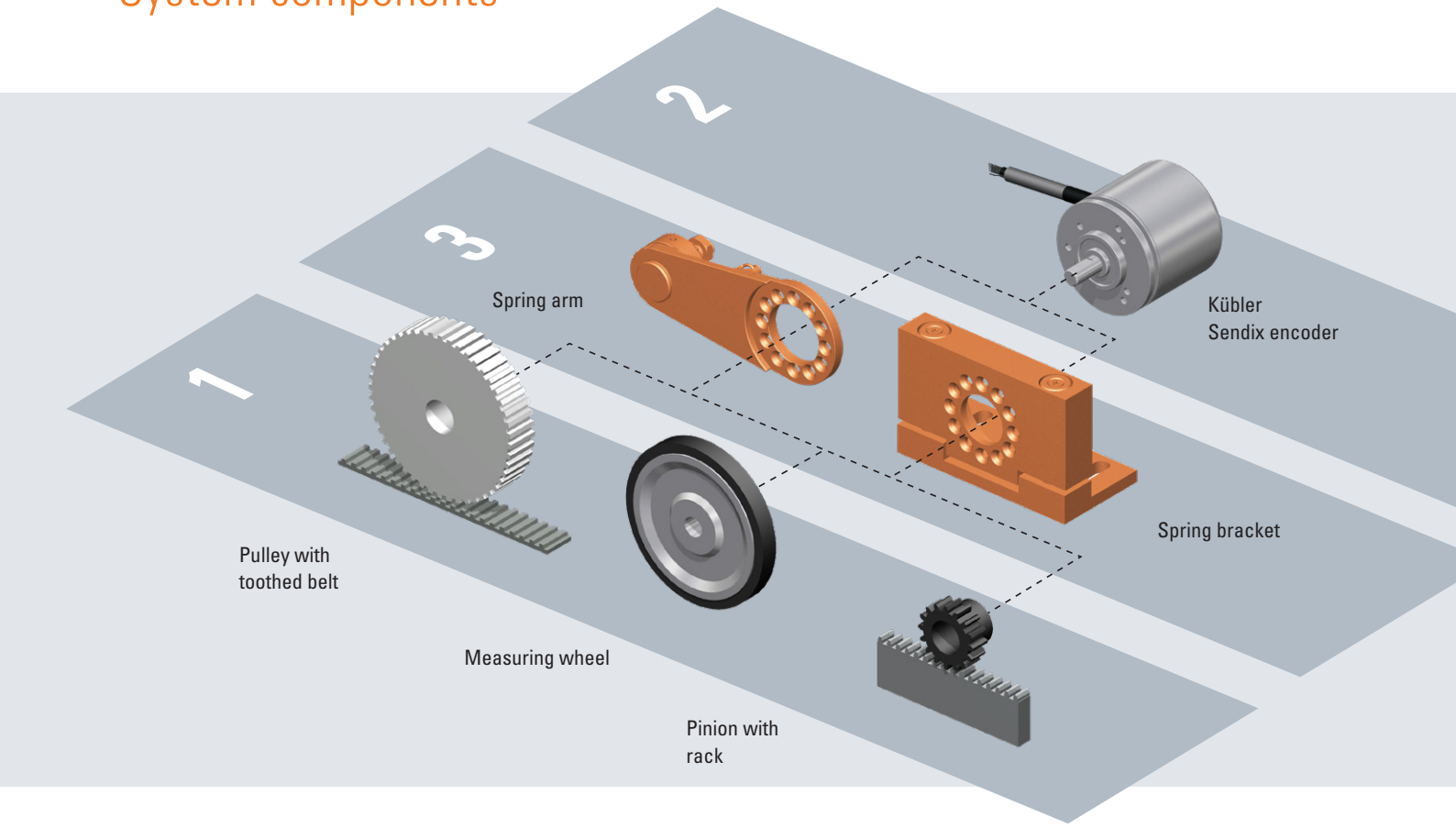
Measuring wheel systems are used for the detection of linear movements

The linear movement (a) of a material to be measured is converted into a rotational movement (b) by a contacting measuring wheel (or pulley or pinion). An mounted incremental or absolute encoder (c) detects this rotation and converts it into speed, position and distance values.

The linear resolution of the measuring wheel system (mm or inch) results from the ratio of the measuring wheel circumference and the rotary resolution of the encoder (ppr).

	Measuring step (distance/pulse)			Resolution (pulses/distance)		
Calculation	$\frac{\text{distance}}{\text{ppr}}$	=	$\frac{\text{Measuring wheel circumference}}{\text{Pulse number encoder}}$	$\frac{\text{ppr}}{\text{distance}}$	=	$\frac{\text{Pulse number encoder}}{\text{Measuring wheel circumference}}$
Example 1 Measuring wheel circumference = 300 mm Pulse number encoder = 3000 ppr	$\frac{300\text{ mm}}{3000\text{ ppr}}$	=	0.1 mm / puls	$\frac{3000\text{ ppr}}{300\text{ mm}}$	=	10 pulses / mm
Example 2 Measuring wheel circumference = 12inch Pulse number encoder = 1200 ppr	$\frac{12\text{ inch}}{1200\text{ ppr}}$	=	0.01 inch / puls	$\frac{1200\text{ ppr}}{12\text{ inch}}$	=	100 pulses / inch

System components



Due to the functional principle, the measuring wheel systems have a 3-part structure

1. Detection of the linear movement

The linear movement is detected by means of a measuring wheel that runs directly on the material to be measured.

Different measuring wheel coatings
Measuring wheels with different coatings are available for different surfaces of the material to be measured. They ensure slip-free contact between the material to be measured and the measuring wheel.

Different measuring wheel circumferences
The interaction of encoder resolution and measuring wheel circumference is decisive for the calculation of the measurement result.

Special versions
Systems with a combination of pinion/rack or pulley/toothed belt offer 100 % measuring accuracy without slip.

2. Data detection / data transfer

The measuring wheels are connected directly to the shaft of a Kübler Sendix encoder.
Incremental encoders, encoders for all relevant fieldbus systems and versions for Industry 4.0 concepts - the Kübler portfolio offers the right solution for every network requirement.

3. Spring-loaded encoder mounting

Spring arms and spring brackets are available in different sizes and with different contact forces for different applications.
Integrated springs press the measuring wheel onto the surface of the material to be measured. This enables slip-free measurement and also compensates for tolerances perpendicular to the movement of the material being measured.

Portfolio overview measuring wheel systems

				
		Compact-Line		
Type		**MWE11**	**MWE21**	**MWE31**
Design spring element		Spring bracket	Spring arm MWE20	Spring bracket MWE30
Highlights		Smallest size	Adjustable preload Flexible mounting options	Compact design Internal springs
Measuring wheel circumference	recommended	100 mm	200 mm / 6“	200 mm
	other options	–	–	–
Contact force max.		10 N	25 N	15 N
Spring travel max.		10 mm	50 mm	10 mm
Encoder	clamping flange / shaft	ø 24 mm / ø 6 mm	ø 36 or 40 mm / ø 6 mm	ø 36 or 40 mm / ø 6 mm
Encoder interfaces incremental		Push-Pull HTL	Push-Pull HTL; RS422 TTL; Open Collector NPN	Push-Pull HTL; RS422 TTL; Open Collector NPN
Encoder interfaces absolute		–	Analog output; SSI; CANopen; IO-Link	Analog output; SSI; CANopen; IO-Link



further encoders / interfaces on request







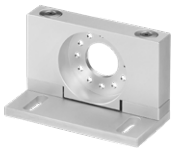
Performance-Line

MWE21.7 with flange adapter	MWE41	MWE61	MWE62 Double measuring wheel system
Spring arm MWE20	Spring bracket MWE40	Spring arm MWE60	
Compact spring arm with encoders in sizes 50 or 58 mm for higher performance	Compact design Internal springs	Highest contact force, adjustable	
300 mm / 12“ 200 mm, 500 mm	300 mm 12“	300 mm / 12“ 200 mm, 500 mm	
25 N	25 N	40 N	
50 mm	10 mm	80 mm	
ø 58 mm / ø 10 mm	ø 58 mm / ø 10 mm	ø 58 mm / ø 10 mm	
Push-Pull HTL · RS422 TTL · Open Collector NPN	Push-Pull HTL · RS422 TTL · Open Collector NPN	Push-Pull HTL · RS422 TTL · Open Collector NPN	
Analog output SSI CANopen IO-Link PROFINET EtherNet/IP PROFIBUS	Analog output SSI CANopen IO-Link PROFINET EtherNet/IP PROFIBUS	Analog output SSI CANopen IO-Link PROFINET EtherNet/IP PROFIBUS	–

Portfolio overview system components

Spring arms / spring brackets





		Compact-Line	
Type		**MWE20** Spring arm	**MWE30** Spring bracket
Highlights		Adjustable contact force Flexible mounting options	Compact design Internal springs
For encoders	clamping flange / shaft	ø 36 or 40 mm / ø 6 mm	ø 36 or 40 mm / ø 6 mm
Measuring wheel circumference	recommended	200 mm / 6“	200 mm
	other options	–	–
Contact force max.		25 N	15 N
Spring travel max.		50 mm	10 mm

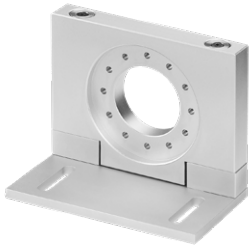
Measuring wheels

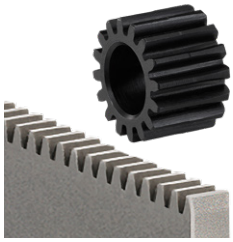
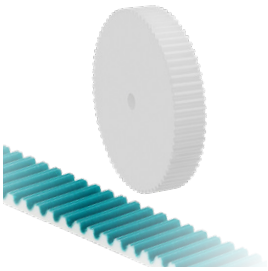






		Diamond knurl (aluminum)	**Rubber / Plastic smooth** (polyurethane)	**Tufted rubber** (polyurethane)
Coating measuring wheel				
Measured material surface / Application	Cardboard, wood, textile	●	●	●
	Plastic, paper	●	●	●
	Wire, greased metals, steel profiles, leather	–	●	–
	Carpet, cables, nonwoven	–	–	●
	Greased metals, glass, floor coverings	–	–	–
	Painted surfaces	–	●	–
	Rubber, soft plastic	●	–	–
Measuring wheel circumference (Width of the tread)	100 mm (9,5 mm)	●	●	–
	200 mm (5,5 mm)	●	●	–
	200 mm (12 mm)	●	●	●
	300 mm (12 mm)	●	●	●
	500 mm (25 mm)	●	●	●
	6 “ (5,5 mm)	●	●	–
	6 “ (12 mm)	●	–	–
	12 “ (12 mm)	●	●	●

		
Performance-Line		
MWE20 + flange adapter Spring arm with flange adapter 8.0000.7000.0079	**MWE40** Spring bracket	**MWE60** Spring arm
Combining the compact design of the spring arm with the higher performance of encoders in sizes 50 or 58 mm	Compact design Internal springs	Highest contact force, adjustable
ø 58 mm / ø 10 mm	ø 58 mm / ø 10 mm	ø 58 mm / ø 10 mm
300 mm / 12“ 200 mm, 500 mm	300 mm 12“	300 mm / 12“ 200 mm, 500 mm
25 N	25 N	40 N
50 mm	10 mm	20 mm

				
O-ring (NBR70)	**Double O-ring** (NBR70)	**Plastic corrugated** (polyurethane)	**Pinion** **+ rack**	**Belt pulley** **+ Toothed belt**
•	•	•	· 100 % slip-free · For encoder shaft 6 mm and 10 mm · Rack 1 m, stackable	· 100 % slip-free · For encoder size ø 58 mm · Toothed belt up to 100 m
•	•	•		
–	–	–		
–	–	–		
–	–	•		
•	•	•		
–	–	–		
–	–	–	Pitch circumference = 50 mm	Pitch circumference = 360 mm
•	–	–		
•	•	•		
•	•	•		
–	–	•		
•	–	–		
–	–	–		
•	•	•		

Portfolio overview Kübler Sendix encoders

Kübler Sendix encoders (recommended selection - for further variants see short form catalog)						
	incremental					
Technology	optical sensor					
Type	**2400**	**KIS40**	**KIS50.8D** double-sided shaft	**KIS50**	**5805**	**M3661**
Interface	Push-Pull HTL	Push-Pull HTL RS422 TTL Open Collector NPN	Push-Pull HTL RS422 TTL Open Collector NPN	Push-Pull HTL RS422 TTL Open Collector NPN	Push-Pull HTL RS422 TTL	Analog output
Resolution max.	1.024 ppr	2.500 ppr		5.000 ppr	36.000 ppr	12 bit ST 16 bit MT
Suitable for measuring wheel system	**MWE11**	**MWE21** **MWE31**	**MWE62**	**MWE41** **MWE61**		
Clamping flange	ø 24 mm	ø 40 mm	ø 58 mm	ø 58 mm	ø 58 mm	ø 36 mm
Shaft diameter	6 mm	6 mm	10 mm	10 mm	10 mm	6 mm

Industry 4.0 / IIoT ready



„Industry 4.0 / IIoT ready" means: In addition to the classical measuring task and transmission of measured values, encoders must also provide further functionalities for networking the products and collecting or transmitting additional information.
Which Industry 4.0 / IIoT functionalities an encoder needs to be „Industry 4.0 / IIoT ready" depends on the overall concept. The decisive factor here is the role assigned to the encoder. Either as a part of or as an independent Industry 4.0 / IIoT object (asset). This determines whether the encoder must have its own administration shell or be integrated into an existing administration shell.

kuebler.com/iiot

PROFINET · EtherCAT Technology Group · EtherNet/IP · ETHERNET POWERLINK · CC-Link IE · OPC UA · TSN · IO-Link · SCS open link

							
absolute multiturn							
magnetic sensor					optical sensor		mechanical drive
M3663	**M3668**	**M5861**	**M5863**	**M5868**	**F5863**	**F5868**	**5868**
SSI	CANopen IO-Link	Analog output	SSI	CANopen IO-Link	SSI BiSS INTERFACE	CANopen PROFINET Modbus EtherNet/IP	CANopen PROFINET PROFIBUS EtherCAT
14 bit ST 24 bit MT	14 bit ST 29 bit MT	12 bit ST 16 bit MT	14 bit ST 24 bit MT	14 bit ST 29 bit MT	17 bit ST 24 bit MT	16 bit ST 16 bit MT	16 bit ST 12 bit MT
MWE21 **MWE31**		**MWE41** **MWE61**					
ø 36 mm	ø 36 mm	ø 58 mm	ø 58 mm	ø 58 mm	ø 58 mm	ø 58 mm	ø 58 mm
6 mm	6 mm	10 mm	10 mm	10 mm	10 mm	10 mm	10 mm

IO-Link

IO-Link is establishing itself more and more on the market - and the trend is rising. IO-Link is used today in machine tools, production lines, intralogistics and packaging machines. IO-Link stands for simplicity, cost reduction and as a starting point for implementing future Industrie 4.0 / IIoT concepts. IO-Link products from Kübler open up new possibilities for your application.

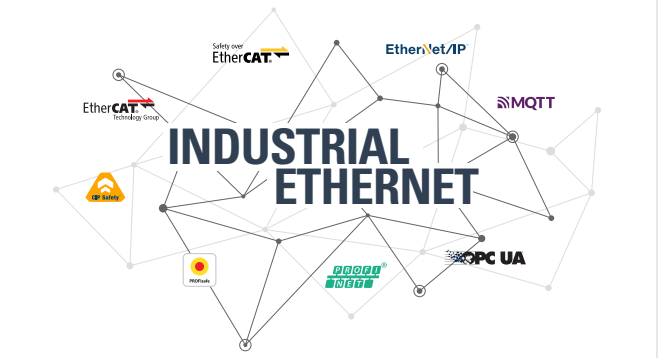
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Industrial Ethernet

The use of Industrial Ethernet communication in modern industry is continuously increasing. In the future, in line with the Industrie 4.0 idea, all areas of industrial production plants will be united in a single network on the Industrial Ethernet platform, from the field devices to the control level to the cloud. And this with real-time data exchange. The corresponding communication capability of the sensors plays an essential role here.

kuebler.com/industrial-ethernet

- **Time and cost savings**
- **Independent in use**
- **Efficient production thanks to Smart Sensor profile**
- **Remote diagnosis and condition monitoring**



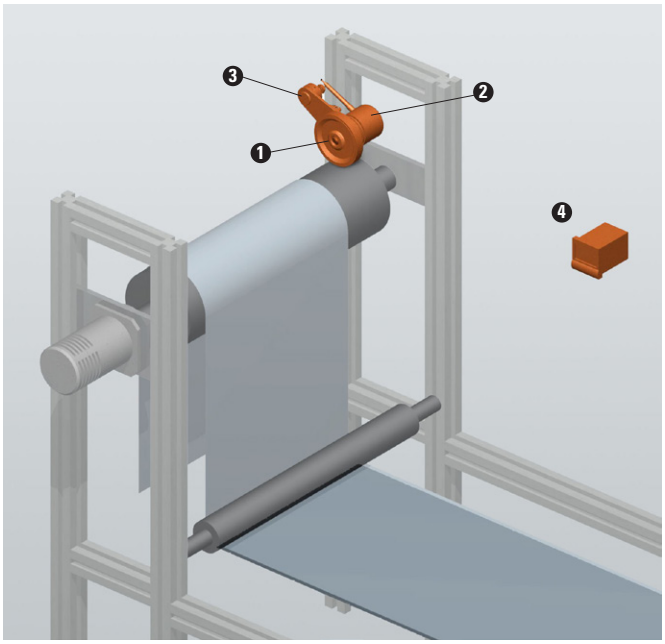
Measuring wheel systems - applications

Measurement of speed, position and distance directly on conveyor belts, steel and sheet metal working machines, storage and conveyor systems, sorting systems, conveyor belts, textile machines, printing and paper industry etc.

Application examples:

Speed measurement on a guide shaft

Requirement
- Reliable speed measurement
- Continuous contact force
- Easy to integrate
- Compact design



Kübler solution
Measuring wheel system **MWE21** consisting of



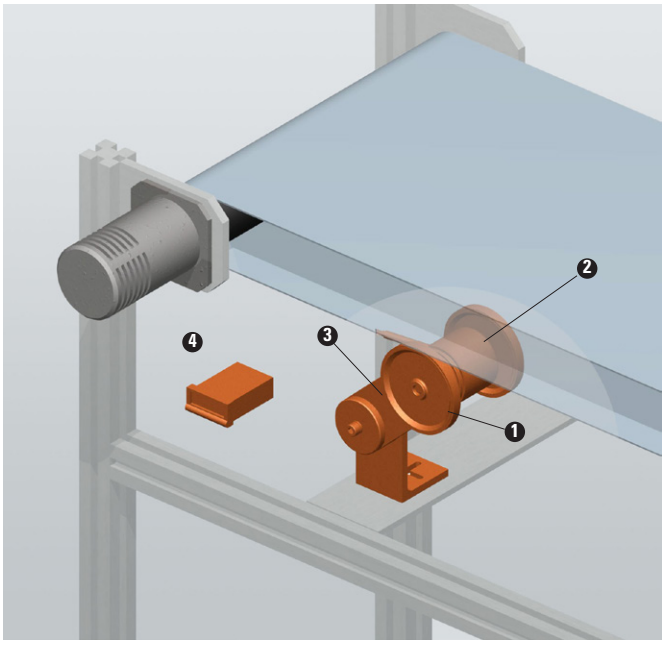
❶	Measuring wheel	circumf. 200 mm, plastic smooth
❷	Encoder	Sendix KIS40, 1000 ppr
❸	Spring arm	MWE20

❹ Preset counter Codix 924
The integrated speedometer function makes it easy to visualize and control speeds.



Speed measurement on a conveyor belt

Requirement
- Direct measurement on the material to be measured (not via a motor encoder -> slip in the system)
- Can be mounted from below against the conveyor belt (overhead mounting)
- High contact pressure
- Large spring travel (tolerance compensation from conveyor belt)
- Redundant double measuring wheel system (two support points for the measuring wheels)



Kübler solution
Measuring wheel system **MWE62** consisting of



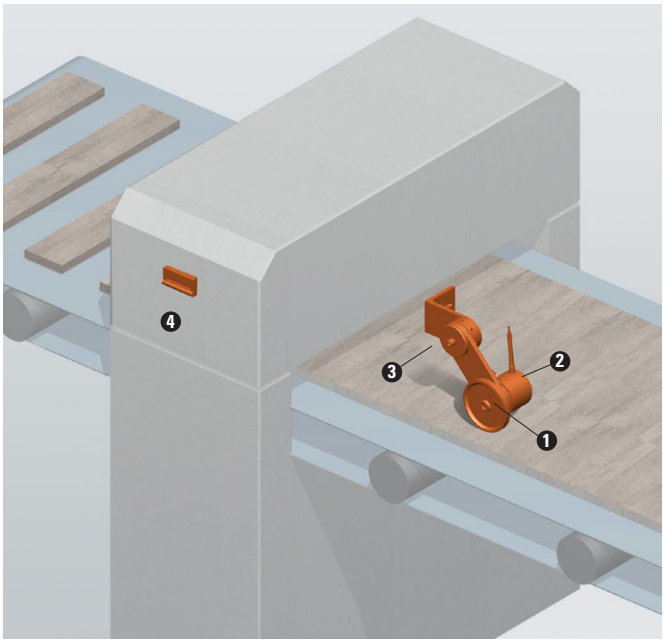
❶	Measuring wheel	circumf. 300 mm, double O-ring
❷	Encoder	Sendix KIS50, 1000 ppr
❸	Spring arm	MWE60

❹ Preset counter Codix 571T
In addition to a touch function, the LED preset counter has the option of visualizing measured values (presets) with color changes when they are reached or fallen short of.



Length measurement - example wooden boards

Requirement
- Precise length measurement
- High contact forces
- Easy maintenance
- Robust overall system



Kübler solution
Measuring wheel system **MWE61** consisting of

❶	Measuring wheel	circumf. 300 mm, diamond knurl
❷	Encoder	Sendix KIS50, 1000 ppr
❸	Spring arm	MWE60



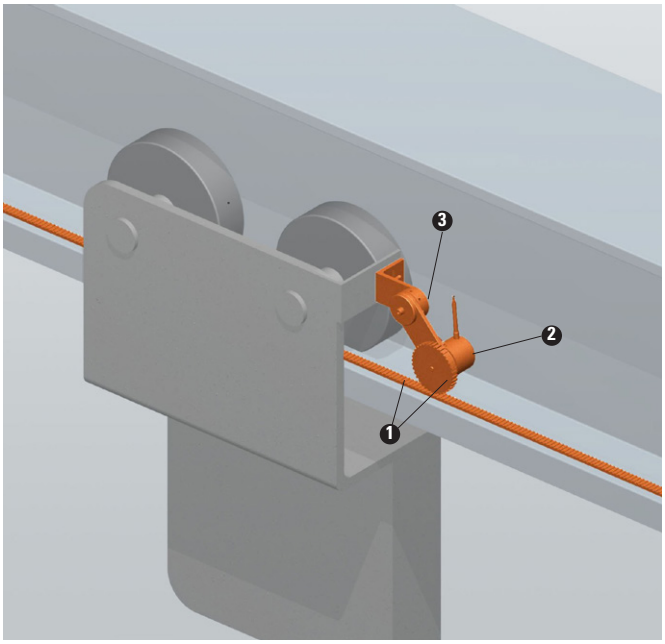
❹ Preset counter Codix 560

The desired measuring length can be set via the LED preset counter. When the set value is reached, a cut-to-length process is triggered.



Position measurement on gantry cranes

Requirement
- 100 % slip-free measurement
- Exact positioning
- Absolute position feedback
- High contact force
- High shock resistance



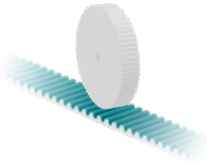


Kübler solution
Measuring wheel system **MWE61** consisting of:

❶	Measuring wheel	Pulley with toothed belt
❷	Encoder	Sendix F5868 PROFINET
❸	Spring arm	MWE60



Direct communication with the controller via PROFINET interface.



Measuring wheel system MWE11
Incremental encoder 2400

The compact measuring wheel system MWE11 with the smallest size can be integrated very flexibly, even in the tightest installation spaces.



- **Easy handling**
  Measuring wheel, sensor and spring bracket are pre-assembled and therefore easy to install: screw on - connect - done.
- **Compact design**
  Dimensions of the complete unit only 74 x 50 x 52 mm.
- **Measuring wheels in 2 variants**
  Circumference 100 mm - measuring wheel coating available with diamond knurl or rubber surface.

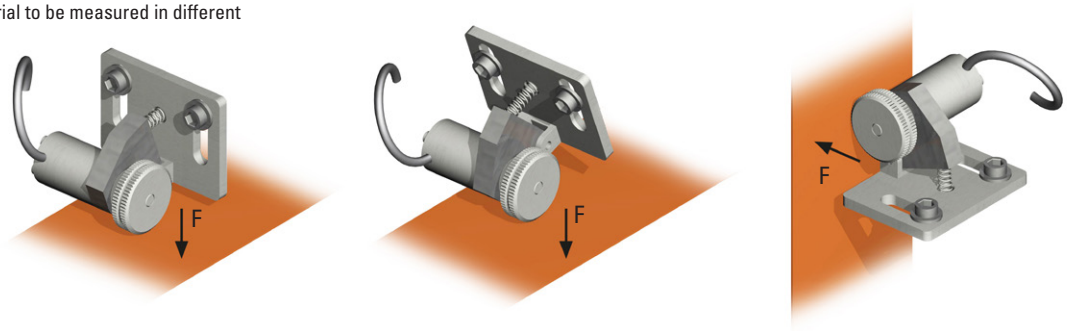
FEATURES AT A GLANCE

- **Contact force max. 10 N**
- **Spring travel max. 10 mm**
- **Measuring wheel circumference 100 mm**
- **Encoder size ø 24 mm**
- **Incremental**

Technology in detail

Various mounting options

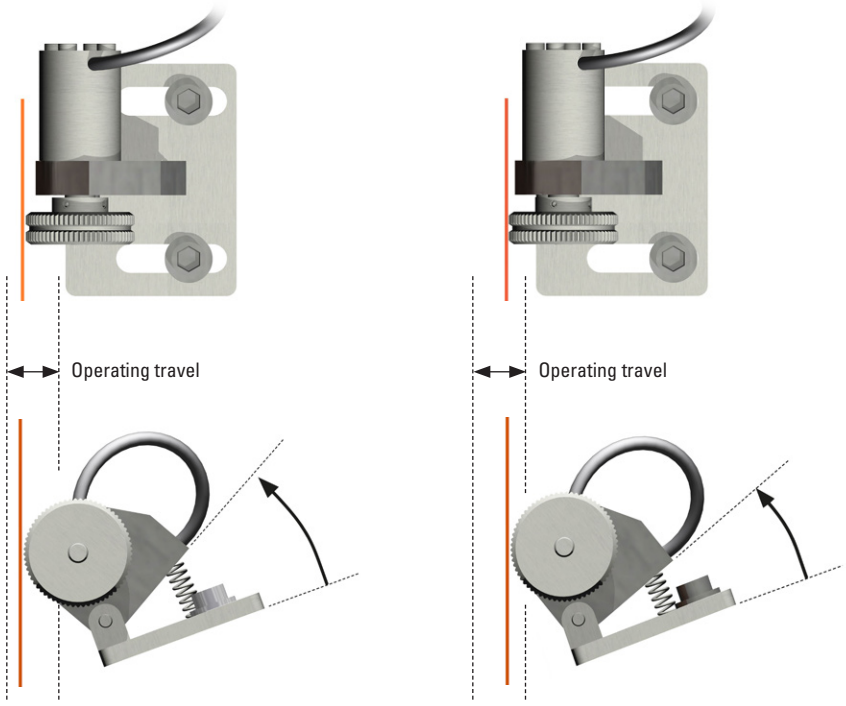
The measuring wheel system can be placed on the material to be measured in different ways.



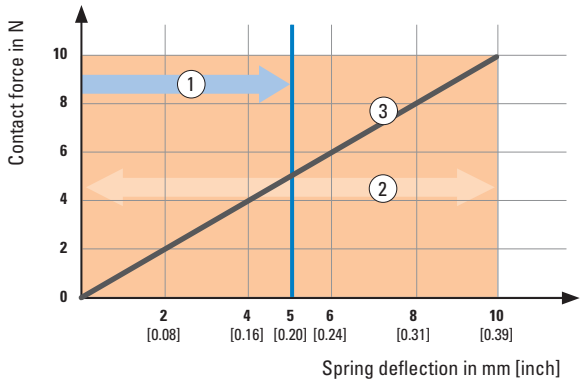
Setting the preload

The distance between the MWE11 measuring wheel system and the material to be measured can be adjusted via 2 slotted holes.

This simultaneously sets the desired preload of the spring.



Contact force of the measuring wheel on the material to be measured



1 Preload, recommended : 5 N

2 Operating travel, max. : 10 mm

3 Contact force in relation to spring deflection

Measuring wheel system MWE21
Incremental or absolute encoders size ø 36 / 40 mm

The compact measuring wheel system MWE21 with adjustable preload can be integrated very flexibly even in the tightest installation spaces.



- **Wide range of encoders**
  Incremental Sendix encoder with a max. resolution of up to 2500 pulses/revolution as well as absolute encoders for different communication interfaces such as IO-Link for integration in Industry 4.0 / IIoT concepts.
- **Suitable measuring wheels for all measured material surfaces**
  Circumference 200 mm or 6" - measuring wheel coating available with O-ring, smooth plastic or diamond knurl surface.
- **Contact force up to max. 25 N**
  With adjustable preload and mechanical spring deflection limitation for a long service life. The integrated spring ensures a working range of the measuring wheel of up to 50 mm vertical to the measuring surface to compensate for tolerances.

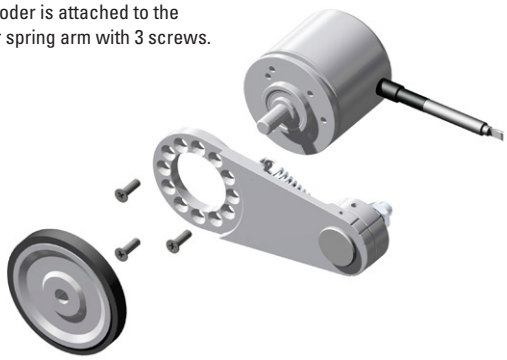
FEATURES AT A GLANCE

- **Contact force max. 25 N**
- **Spring travel max. 50 mm**
- **Measuring wheel circumference 200 mm / 6"**
- **Encoder size ø 36 / 40 mm**
- **Incremental or absolute**

Technology in detail

Mounting options encoder on spring arm

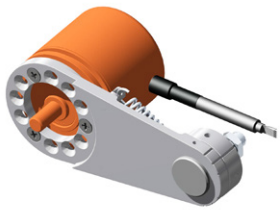
The encoder is attached to the encoder spring arm with 3 screws.



The fastening points are designed in such a way that mounting on both sides of the encoder spring arm is possible.



Mounting left (Delivery state)



Mounting right

For a flexible cable outlet direction, the encoder can additionally be mounted in 30° steps.



0° (delivery state)



30°



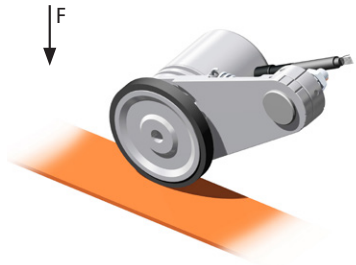
60°



90°

Various mounting options

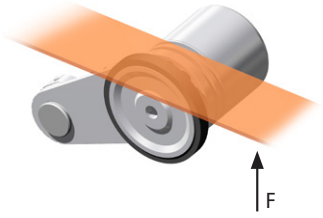
horizontally



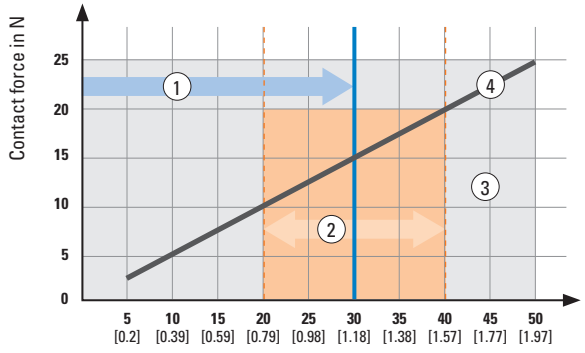
vertically



overhead



Contact force of the measuring wheel on the material to be measured



1	Recommended preload:	15 N (approx. 30 mm defection)
2	Rec. operating travel:	± 10 mm (from the rec. preload)
3	Maximum spring deflection:	50 mm
4	Contact force in relation to spring deflection (Functional principle based on 2 integrated springs)	

Measuring wheel system MWE31

Incremental or absolute encoders size ø 36 / 40 mm

The compact MWE31 measuring wheel system with internal springs can be quickly and easily integrated into even the tightest installation spaces.



- **Wide range of encoders**
  Incremental Sendix encoder with a max. resolution of up to 2500 pulses/revolution as well as absolute encoders for different communication interfaces such as IO-Link for integration in Industry 4.0 / IIoT concepts.
- **Suitable measuring wheels for all measuring surfaces**
  Circumference 200 mm - measuring wheel coating available with O-ring, smooth plastic or diamond knurl surface.
- **Contact force up to max. 15 N**
  The integrated spring ensures a working range of the measuring wheel of up to 10 mm vertical to the measuring surface to compensate for tolerances.

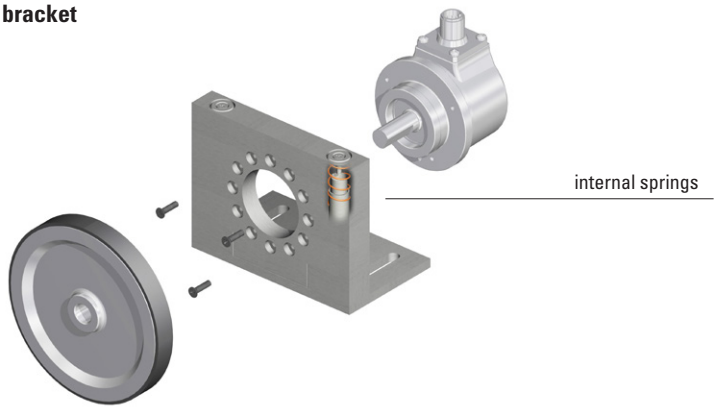
FEATURES AT A GLANCE

- **Contact force max. 15 N**
- **Spring travel max. 10 mm**
- **Measuring wheel circumference 200 mm**
- **Encoder size ø 36 / 40 mm**
- **Incremental or absolute**

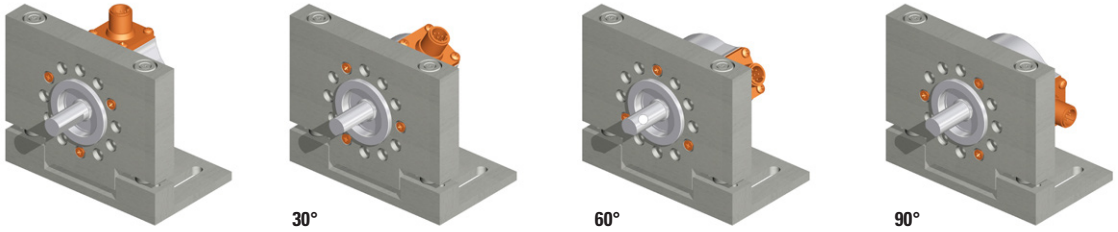
Technology in detail

Mounting options encoder on spring bracket

The encoder is attached to the spring bracket with 3 screws.



For a flexible outlet direction of the cable or connector, the encoder can additionally be mounted in 30° steps.

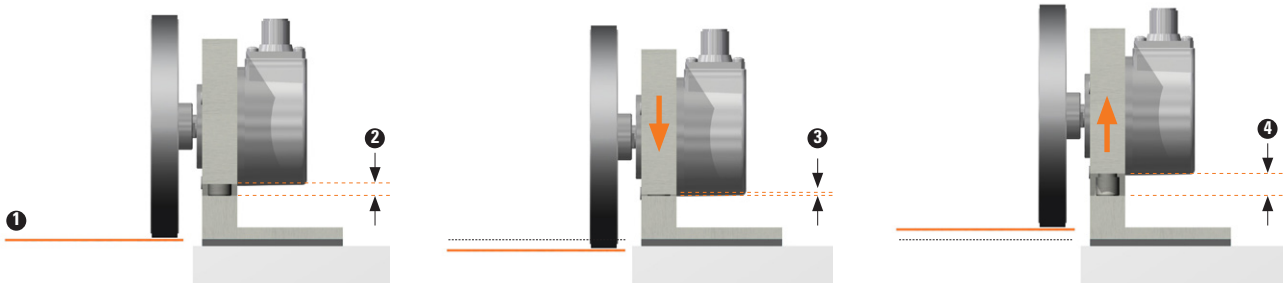


0° (delivery state) **30°** **60°** **90°**

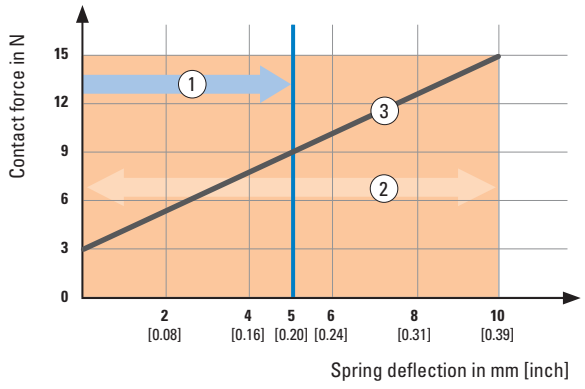
Mounting on the application

Install the MWE31 on the material to be measured ❶ in such a way that the requested preload is obtained. (ideally approx. 5 mm of the spring deflection ❷)

The working range is from 0 mm ❸ (equivalent to 3 N) to 10 mm ❹ (equivalent to 15 N)



Contact force of the measuring wheel on the material to be measured



1 Preload, recommended : 9 N (approx. 5 mm defection)

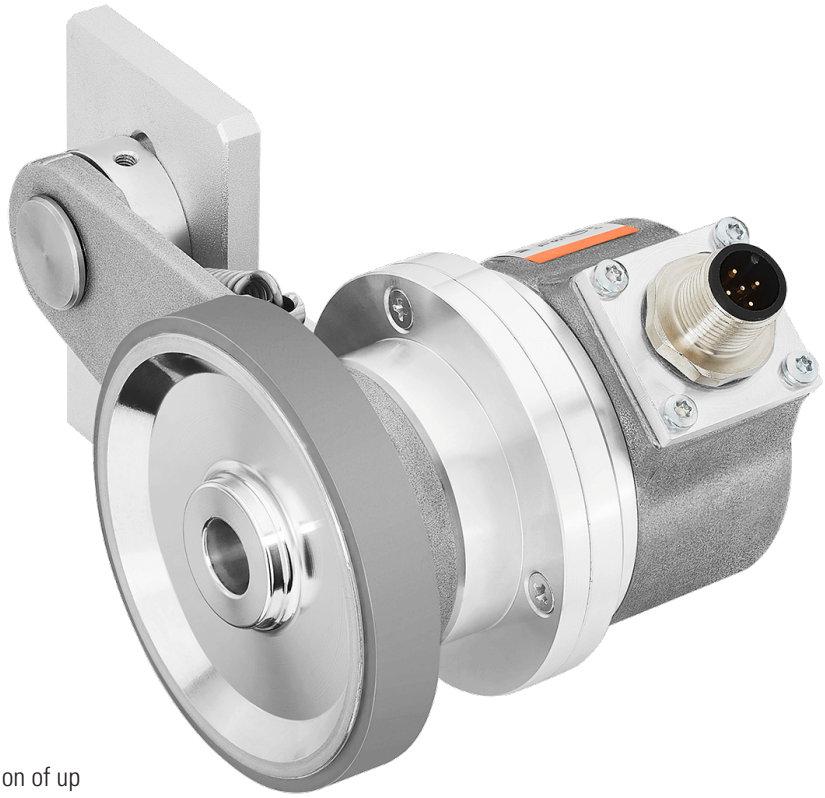
2 Operating travel, max. : 10 mm

3 Contact force in relation to spring deflection

(Functional principle based on 2 integrated springs)

Measuring wheel system MWE21.7 with flange adapter
Incremental or absolute encoders size ø 58 mm

The compact MWE21.7 measuring wheel system with flange adapter combines the compact design of the MWE20 spring arm with the high performance of 58 mm encoders.



- **Wide range of encoders**
  Incremental Sendix encoder with a max. resolution of up to 36.000 pulses/revolution as well as absolute encoders for different communication interfaces such as IO-Link for integration in Industry 4.0 / IIoT concepts.
- **Suitable measuring wheels for all measured material surfaces**
  Circumference 300 mm or 12" – measuring wheel coating available with O-ring, double O-Ring, plastic (smooth, corrugated or tufted rubber) or diamond knurl surface.
- **Contact force up to max. 25 N**
  With adjustable preload and mechanical spring deflection limitation for a long service life. The integrated spring ensures a working range of the measuring wheel of up to 50 mm vertical to the measuring surface to compensate for tolerances.

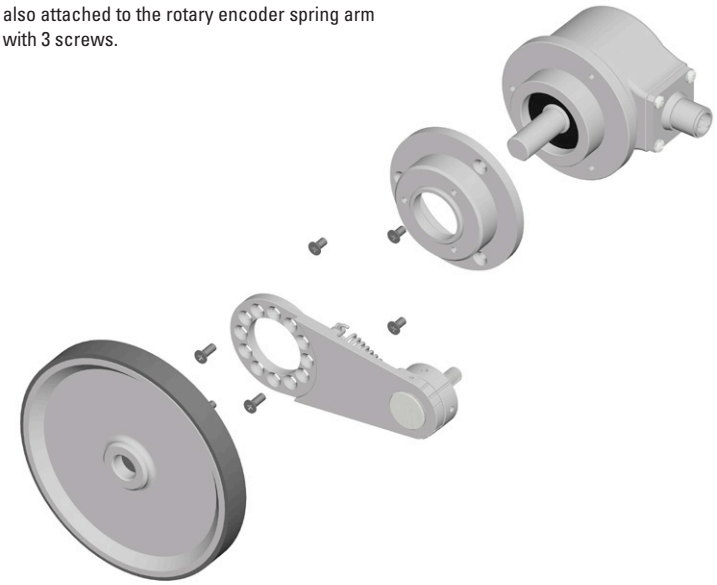
FEATURES AT A GLANCE

- **Contact force max. 25 N**
- **Spring travel max. 50 mm**
- **Measuring wheel circumference 300 mm / 12"**
- **Encoder size ø 58 mm**
- **Incremental or absolute**

Technology in detail

Mounting options encoder on spring arm

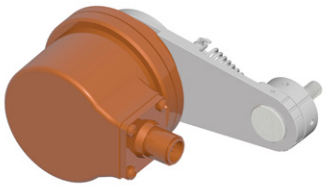
The rotary encoder is attached to the flange adapter with 3 screws, and the flange adapter is also attached to the rotary encoder spring arm with 3 screws.



The fastening points are designed in such a way that mounting on both sides of the encoder spring arm is possible.

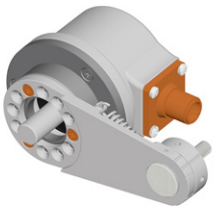


Mounting left (Delivery state)

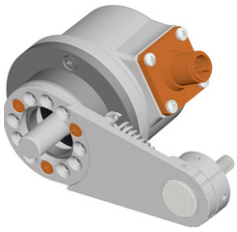


Mounting right

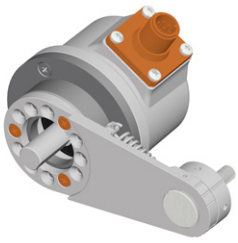
For a flexible cable outlet direction, the encoder can additionally be mounted in 30° steps.



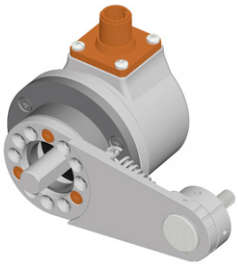
0° (delivery state)



30°

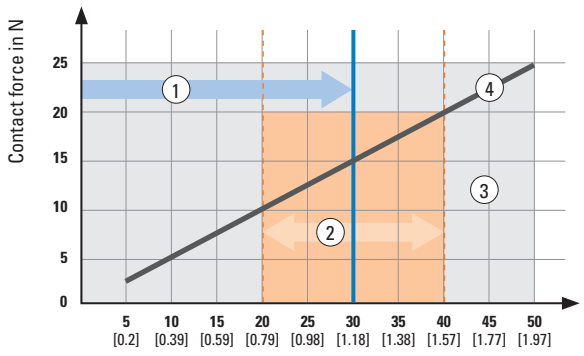


60°



90°

Contact force of the measuring wheel on the material to be measured



1	Recommended preload:	15 N (approx. 30 mm defection)
2	Rec. operating travel:	± 10 mm (from the rec. preload)
3	Maximum spring deflection:	50 mm
4	Contact force in relation to spring deflection (Functional principle based on 2 integrated springs)	

Measuring wheel system MWE41
Incremental or absolute encoders size ø 58 mm

The MWE41 measuring wheel system with internal springs can be quickly and easily integrated into many applications.



- **Wide range of encoders**
  Incremental Sendix encoders with a max. resolution of up to 36,000 pulses/revolution as well as absolute encoders for different communication interfaces such as IO-Link or Profinet for integration in Industry 4.0 / IIoT concepts.
- **Suitable measuring wheels for all measuring surfaces**
  Circumference 300 mm – measuring wheel coating available with O-ring, double O-Ring, plastic (smooth, corrugated or tufted rubber) or diamond knurl surface.
- **Contact force up to max. 25 N**
  The internal spring ensures a working range of the measuring wheel of up to 10 mm vertical to the measuring surface to compensate for tolerances.

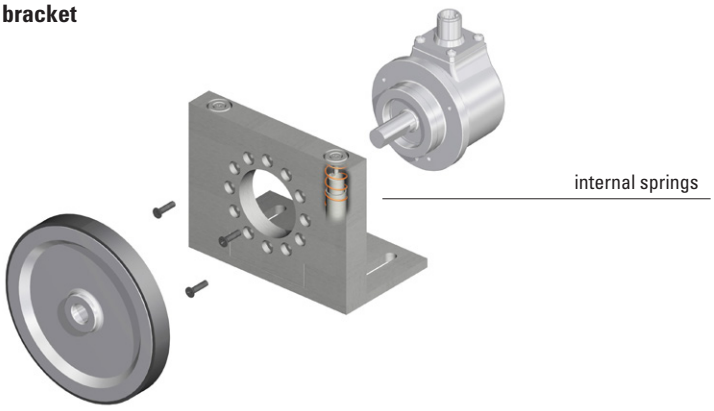
FEATURES AT A GLANCE

- **Contact force max. 25 N**
- **Spring travel max. 10 mm**
- **Measuring wheel circumference 300 mm**
- **Encoder size ø 58 mm**
- **Incremental or absolute**

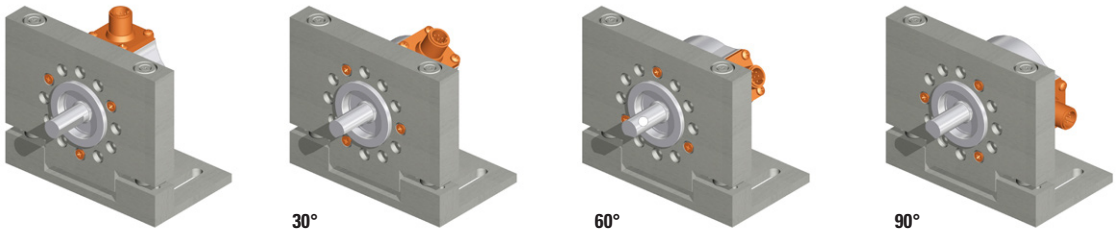
Technology in detail

Mounting options encoder on spring bracket

The encoder is attached to the spring bracket with 3 screws.



For a flexible outlet direction of the cable or connector, the encoder can additionally be mounted in 30° steps.

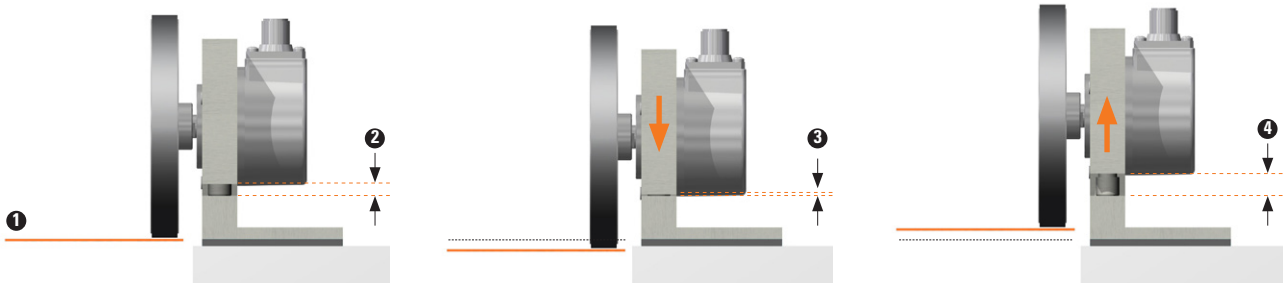


0° (delivery state) **30°** **60°** **90°**

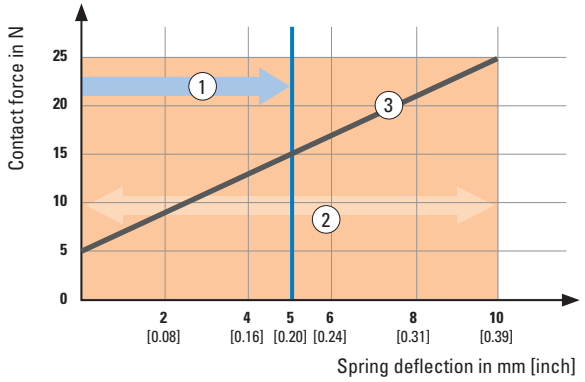
Mounting on the application

Install the MWE41 on the material to be measured ❶ in such a way that the requested preload is obtained. (ideally approx. 5 mm of the spring deflection ❷)

The working range is from 0 mm ❸ (equivalent to 5 N) to 10 mm ❹ (equivalent to 25 N)



Contact force of the measuring wheel on the material to be measured



① Preload, recommended : 15 N (approx. 5 mm defection)

② Operating travel, max. : 10 mm

③ Contact force in relation to spring deflection

(Functional principle based on 2 integrated springs)

Measuring wheel system MWE61
Incremental or absolute encoders size ø 58 mm

The robust MWE61 measuring wheel system offers maximum spring deflection at maximum contact force to compensate for tolerances vertical to the transport movement of the material to be measured.



- **Wide range of encoders**
  Incremental Sendix encoders with a max. resolution of up to 36,000 pulses/revolution as well as absolute encoders for different communication interfaces such as IO-Link or Profinet for integration in Industry 4.0 / IIoT concepts.
- **Suitable measuring wheels for all measuring surfaces**
  Circumference 300 mm or 12" – measuring wheel coating available with O-ring, double O-Ring, plastic (smooth, corrugated or tufted rubber) or diamond knurl surface.
- **Contact force up to max. 40 N**
  Utilizes a stepless adjustable preload. To compensate for tolerances, the integrated spring ensures a working range of the measuring wheel up to a maximum of 80 mm vertical to the measuring surface.

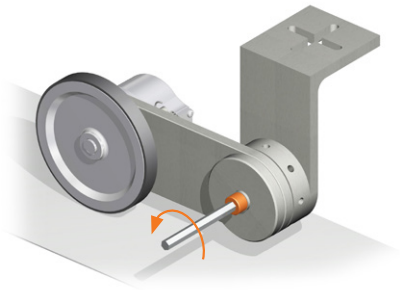
FEATURES AT A GLANCE

- **Contact force max. 40 N**
- **Spring travel max. 80 mm**
- **Measuring wheel circumference 300 mm / 12"**
- **Encoder size ø 58 mm**
- **Incremental or absolute**

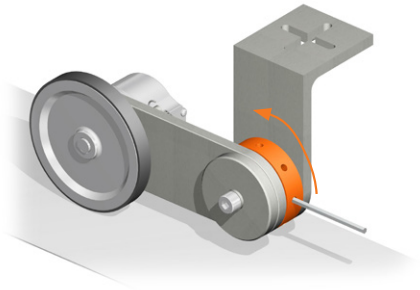
Technology in detail

Setting the preload

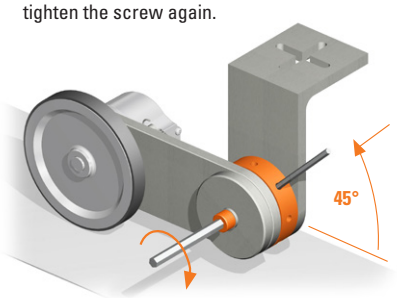
1. Mount the measuring wheel system on the application and release screw.



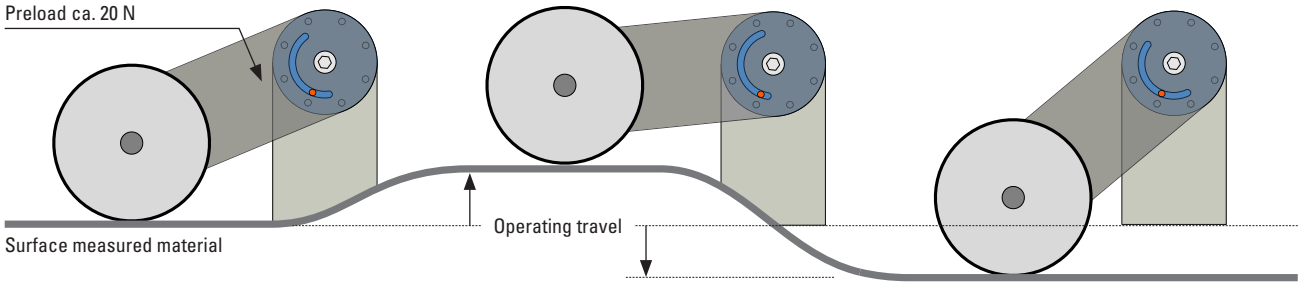
2. Turn the adjustment ring with a thin allen key or screwdriver until the desired preload is reached.



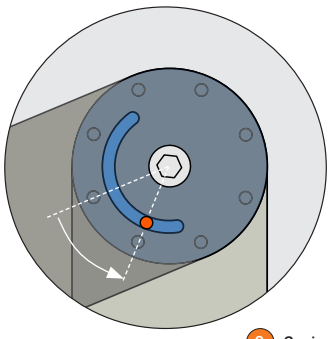
3. As a guide: Internal detent points in 45° steps correspond to approx. 20 N.
Hold the position of the adjustment ring and tighten the screw again.



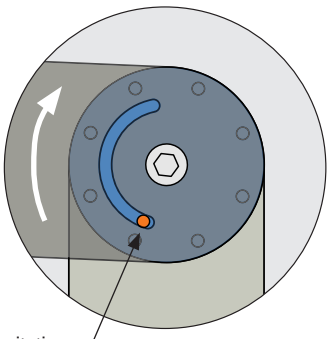
Installation example



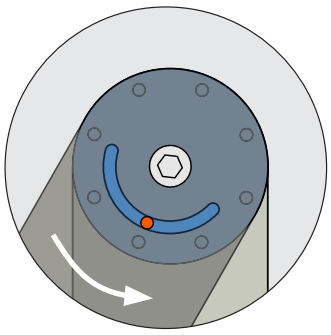
Preload



Contact force max.

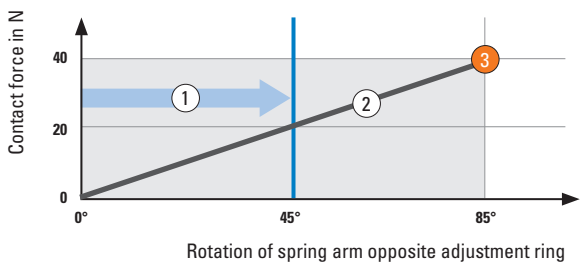


Contact force min.



3 Spring deflection limitation

Contact force of the measuring wheel on the material to be measured



1 Preload (example): 20 N
by turning the setting wheel by approx. 45°
- corresponds to a detent point

2 Contact force

3 Spring deflection limitation to protect against overload

Double measuring wheel system MWE62
Incremental encoder KIS50 (double-sided shaft)

The robust MWE62 measuring wheel system offers maximum spring deflection at maximum contact force to compensate for tolerances perpendicular to the transport movement of the material to be measured.
The use of 2 measuring wheels guarantees optimum contact with the material to be measured, even under difficult conditions.



- **High contact reliability to the measured material**
  The use of a second measuring wheel on the encoder ensures a high degree of contact with the measuring surface even under difficult conditions - high vibrations or unevenness.
- **Suitable measuring wheels for all measuring surfaces**
  Circumference 300 mm or 12“ – measuring wheel coating available with O-ring, double O-Ring, plastic (smooth, corrugated or tufted rubber) or diamond knurl surface.
- **Contact force up to max. 40 N**
  Utilizes a stepless adjustable preload. To compensate for tolerances, the integrated spring ensures a working range of the measuring wheel up to a maximum of 80 mm vertical to the measuring surface.

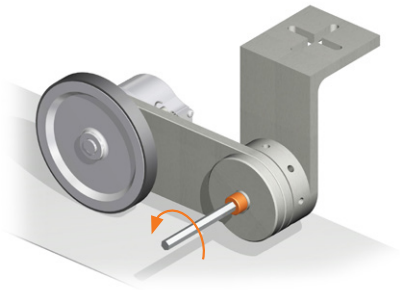
FEATURES AT A GLANCE

- **Contact force max. 40 N**
- **Spring travel max. 80 mm**
- **Measuring wheel circumference 300 mm / 12“**
- **Encoder size ø 58 mm**
- **Incremental**

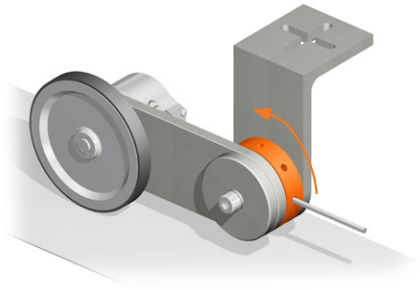
Technology in detail

Setting the preload

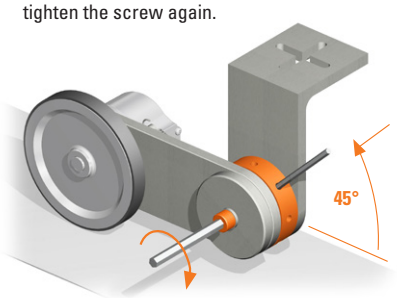
1. Mount the measuring wheel system on the application and release screw.



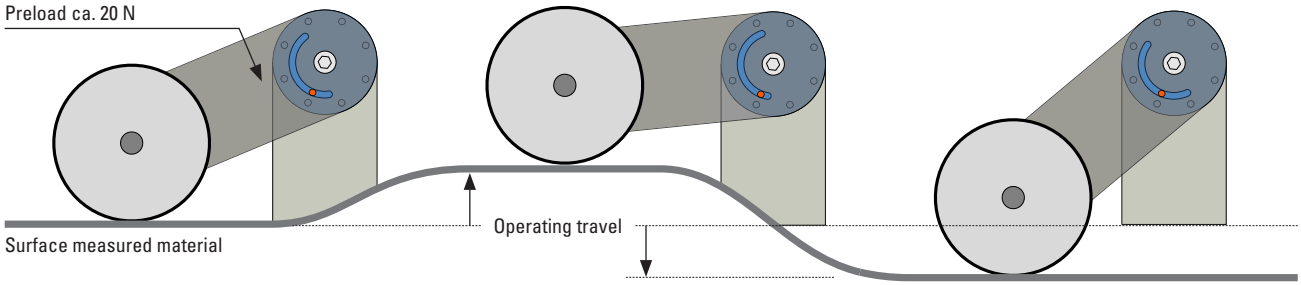
2. Turn the adjustment ring with a thin allen key or screwdriver until the desired preload is reached.



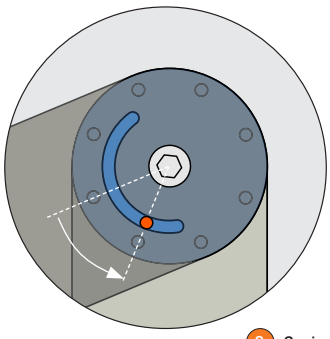
3. As a guide: Internal detent points in 45° steps correspond to approx. 20 N.
Hold the position of the adjustment ring and tighten the screw again.



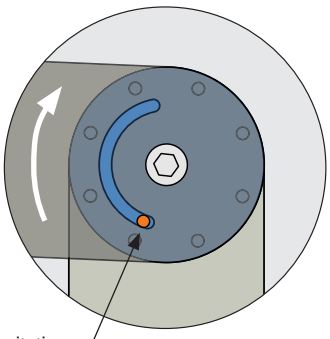
Installation example



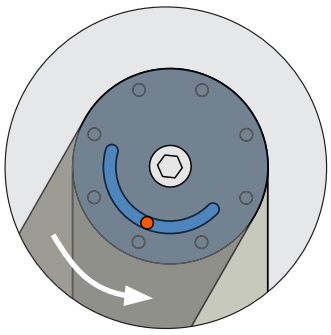
Preload



Contact force max.

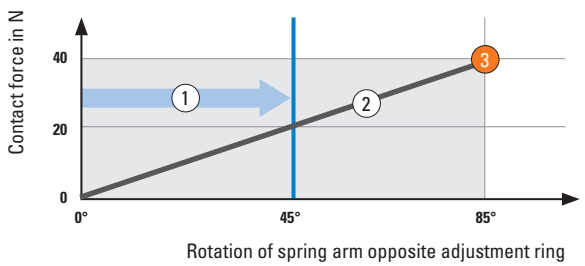


Contact force min.



3 Spring deflection limitation

Contact force of the measuring wheel on the material to be measured



1 Preload (example): 20 N
by turning the setting wheel by approx. 45°
- corresponds to a detent point

2 Contact force

3 Spring deflection limitation to protect against overload



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